Lanka Data: A Four-Field Grammar for Querying Public Data about Sri Lanka

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Abstract

Lanka Data is a software library that provides a single, uniform interface to public data about Sri Lanka. Rather than exposing a growing collection of endpoints, methods, and parameters, it reduces every query to one string of four positional fields—*what* is measured, *when*, *where*, and *how* it is presented—delimited by slashes. The same string serves unchanged as a Python argument, a command-line argument, a URL path, and a file path. This paper describes the motivation for the design, specifies the four-field command grammar and its single intentional coupling, argues that the grammar spans the target query space by composition rather than by accretion, and catalogues the datasets— census, election, administrative geography, and hydrology—and their sources that the library exposes.

1 Introduction

The goal of Lanka Data is "one API to rule them all": a single interface that can express *any* query, rather than a proliferation of endpoints, methods, libraries, and parameter sets that each answer one narrow question. Most data libraries grow by accretion; every new question adds another function, endpoint, or flag, and no single mental model survives contact with the result. We wanted the opposite: a fixed, minimal grammar that a user learns *once* and can then aim at anything, and that a non-technical user can read and write without learning to program. The current domain is public Sri Lankan data— census measurements, election results, and administrative geography—but nothing in the grammar is specific to Sri Lanka: *what*, *when*, *where*, and *how* are the dimensions of essentially any factual query about the world.

2 The Command Grammar

The public interface is a single string parsed into four positional fields, delimited by slashes:

`What / When / Where / How`

There is no secondary configuration surface: no options object, no builder API, no config files. Each field is an independent axis of the query, and the value chosen for one field does not constrain the valid values of another, with a single intentional coupling described below.

2.1 What — the measurement

What identifies the quantity being retrieved. It is a measurement, independent of time, region, and presentation. Measurements include census quantities and election results. A reserved keyword denotes the *absence* of a measurement: a request for region geometry with no data bound to it, used for base maps and for inspecting boundary changes in isolation. Because **What** encodes only the measurement type, a single measurement is reused across every combination of the other three fields without expanding the vocabulary.

2.2 When — the observation time

When binds the measurement to the point or interval in time at which it happened. It accepts either a single year or an interval. An interval is a single value, not two concatenated queries.

2.3 Where — the region

Where identifies the region under measurement: its identity within the administrative hierarchy, not merely a coordinate. It carries the most syntax of the four fields because it is the axis along which real queries vary most. The supported forms are a single region, resolution into child regions of a given type, an explicit set of named regions, a contiguous range between two endpoints, and a historical boundary variant that selects a region's boundaries as they existed before a given boundary redesign. Observation time and boundary epoch are kept as separate values so that counts are not misattributed to the wrong geometry.

2.4 How — the presentation

How specifies the output representation. The same measurement can be emitted as a map, a chart, or raw data without any change to **What**. Because presentation is separated from measurement, **How** carries its own modifier grammar: rankings of a category, shares, differences between observations, and diversity indices are transformations applied to the same underlying values. This is the single intentional coupling of the grammar: change-based modifiers are valid only when **When** supplies an interval.

3 Datasets and Sources

The **What** vocabulary is organised into four data categories. *census-population* exposes measurements such as ethnicity, religion, age group, education, employment, and literacy; *census-housing* exposes housing characteristics such as construction materials, water, lighting, and toilet facilities; *election* exposes presidential, parliamentary, and local government results and summaries; and *rivers* exposes river length and catchment statistics. Each response carries the provenance of the sources below.

3.1 Census of Population and Housing

The census-population category exposes measurements such as ethnicity, religion, age group, education, employment, and literacy. The census-housing category exposes housing characteristics such as construction materials, water, lighting, and toilet facilities. Data is available from the 2001, 2012, and 2024 census cycles (Department of Census and Statistics, 2024a). All census data is sourced from the Department of Census and Statistics.

3.2 Elections

The election category exposes presidential, parliamentary, and local government results and summaries across multiple election years. This includes vote tallies, seat distributions, and electoral statistics at national, district, and constituency levels. All election data is sourced from the Election Commission of Sri Lanka (2024).

3.3 Administrative Geography

The administrative boundary data includes both current and historical boundary epochs, allowing queries to be anchored to the boundaries as they existed at the time of observation. This includes provinces, districts, divisional secretariat divisions, and Grama Niladhari divisions across different time periods. Data is sourced from the Department of Census and Statistics (2024b).

3.4 Rivers

The rivers category exposes river length and catchment statistics for major waterways in Sri Lanka (Lehner and Grill, 2013). This data includes geometric and hydrological properties of river systems. Data is sourced from HydroRIVERS via the `lk_rivers` project.

References

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